

Nicholas Brar

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EDUCATION

Stony Brook University

Bachelor of Science in Mathematics

Stony Brook, NY

Aug. 2021 – May 2025

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, SQL (Postgres), R

ML Frameworks/Libraries: Scikit-learn, TensorFlow, PyTorch, NLP (NLTK, spaCy, Hugging Face)

Developer Tools: Git, Supabase, BitBucket, Jira, Firebase

EXPERIENCE

Climate Data & Modeling Intern

EcoDataLab

Sept 2025 – Present

Berkeley, CA (Remote)

- Redesigned the Consumption Based Emissions Inventory (CBEI) modeling pipeline using CEX data, replacing a single model with **32 annual models** and improving average predictive accuracy (R^2) by **0.08**
- Built a unified React map component that replaced iframe pages, cut load times by **5×**, and enabled fully interactive maps for emissions by tract
- Built an end-to-end R workflow combining supply-chain and economic data to compute and visualize **national non-GHG impact factors** at the local level
- Automated the conversion of high-level impact data into detailed sector-level outputs, enabling more precise and locally relevant analysis

Software Engineer Intern

Dealer Shock

Jan 2025 – May 2025

Cleveland, OH (Remote)

- Built the company's official **React Native** mobile application from scratch using **Expo**, pushing development **12 months** ahead of its original estimated release date
- Automated VIN scanning for clients, reducing manual entry by **90%**
- Collaborated in 2-week sprints using Agile methodologies to deliver iterative improvements

DRP Undergraduate Scholar

Stony Brook University Mathematics Department

Dec 2024 – May 2025

Stony Brook, NY

- Built a **Python**-based MNIST classification pipeline using topological features, surpassing prior topological data analysis (TDA) benchmarks
- Applied modern software design patterns to parallelize feature extraction across **13** branches to extract rich and discriminative feature sets
- TDA publication showcased on Stony Brook's DRP site - [Link to Paper](#)

PROJECTS

Fast American Option Pricing | *Python, C++, Quantitative Finance, Numerical Methods*

- Implemented a high-performance spectral collocation method for American option pricing in C++, based on Jacobi–Newton iteration, Gauss–Legendre quadrature, and Chebyshev polynomial interpolation
- Reproduced results from research literature, achieving computational throughput of 100,000 option prices per second per CPU in the Black–Scholes model

Vehicle Miles Traveled (VMT) Model | *Python, Pandas, NumPy, scikit-learn, Random Forest*

- Built an end-to-end data pipeline to clean, integrate, and feature-engineer household, vehicle, and person-level survey data (NHTS) for predictive modeling
- Developed and evaluated regression and ensemble models (Linear Regression, Random Forest) to predict household vehicle miles traveled (VMT), with weighted performance metrics and feature importance analysis

EXTRACURRICULAR

Chess

Stony Brook University & Pittsburgh Chess Club

- Active member of the Stony Brook Chess Club and Pittsburgh Chess Club
- Led Stony Brook's team to a **2nd place finish** in the Collegiate Chess League (Fall 2024)